NeonPine – manufacturing package

1. Fabrication outputs (`fab/`, H-bridge + USB board, KiCad 10, regenerated by `tools/docs.py`)

File	Content
`NeonPine-gerbers.zip`	Gerber X2 (F/B Cu, Mask, Silkscreen, Paste, Edge.Cuts) + Excellon drill + drill map + job file
`NeonPine-pos.csv`	Pick & place, both sides, mm (front: lamps, light sensor Q2, touch pad; everything else SMD on the back)
`NeonPine-bom.csv`	Grouped BOM (value, footprint, qty). **Add MPN / LCSC columns before quoting**
`NeonPine.step`	3D assembly (lamp and inductor models from `NeonPine.3dshapes/`)
`NeonPine-cover-stand-gerbers.zip`	Rear cover + stand base (one file, two outlines): Edge.Cuts, mask, silkscreen (markings), NPTH drill. **No copper.**

Regenerate:

```
kicad-cli pcb export gerbers -o fab/gerber/ --layers
F.Cu,B.Cu,F.Mask,B.Mask,F.Silkscreen,B.Silkscreen,F.Paste,B.Paste,Edge.Cuts NeonPine.kicad_pcb
kicad-cli pcb export drill -o fab/gerber/ --format excellon --generate-map --map-format gerberx2
NeonPine.kicad_pcb
kicad-cli pcb export pos -o fab/NeonPine-pos.csv --format csv --units mm --side both --exclude-dnp
NeonPine.kicad_pcb
kicad-cli sch export bom -o fab/NeonPine-bom.csv --fields
'Reference,Value,Footprint,${QUANTITY},${DNP}' --group-by Value,Footprint NeonPine.kicad_sch
kicad-cli pcb export step -o fab/NeonPine.step --subst-models --force NeonPine.kicad_pcb
python3 tools/cover.py   # then: kicad-cli pcb export gerbers -o fab/cover/ --layers
F.Mask,B.Mask,F.Silkscreen,Edge.Cuts NeonPine-cover.kicad_pcb
```

2. PCB specification

Parameter	Value
Layers / material	2, FR-4 TG≥135, **UL 94 V-0** (ask the fab for the UL file number for the technical file)
Thickness / copper	1.6 mm / 1 oz (35 µm) both sides
Size	54.8 × 69.1 mm incl. 2 stand tabs, irregular outline (star / tiers), corners R 1.5 mm, routed; 3 × Ø2.2 NPTH
Min track / space	0.15 / 0.15 mm (default class 0.2 mm); HV and HVBUS classes 0.3 mm / **0.5 mm clearance**; 0.2 mm inside the SOT-363 lamp switches
Min drill / via	0.3 mm / 0.6 mm pad (vias tented both sides); NPTH-to-copper 0.19 mm at J1's alignment pegs (GCT land pattern)
Finish	**ENIG** (flat touch pads, nicer look)
Solder mask / silk	green or black mask, white silk, both sides
Panel	breakaway tabs only on the pot bottom edge **between** the stand tabs (keep the tree edges smooth)
Cover + stand PCB	FR-4 1.6 mm, **no copper**, black (or green) mask both sides, white silk on the outer face, Ø2.2 NPTH
Test	100 % electrical test (flying probe)

3. Assembly

1. **Back side SMT** (all SMD parts): solder paste, place, reflow with a SAC305 profile.

2. **Front THT:** light sensor **Q2 TEPT4400**, hand-soldered: the **collector goes in the square pad (pad 1)**, emitter in pad 2. Check the lead identification against the Vishay drawing (the leads are marked E/C), and seat the sensor flat (3 mm ball facing forward). J2 is **not a part**, just three pogo pads on the back for programming.
3. **Lamps (THT, hand or selective solder):**
    - Bend the leads 90° about 1.5 mm from the glass seal, using a bending jig. **Never bend at the glass.**
    - The lamp is symmetric and each lead has its own switch (H-bridge), so either electrode can be made to glow. Fit all lamps the same way round (pad 1 = the left pad seen from the front = K side in the firmware tables) so the "pad-1 electrode" is the same plate on every lamp; turn the electrode pair so both plates are visible from the front.
    - Insert, lay the body flat on the front, solder on the back, and trim the leads to ≤ 1.5 mm (they sit under the rear cover).
    - The star lamp LA1 is mounted standing up through the star.
    - **Fix each lamp with a dab of neutral-cure silicone** under the body (risk H8).
4. **Cleaning: mandatory.** Wash off flux residues (no-clean residue + 105 V + a 2 MΩ feedback divider → leakage and HV drift). Dry at 60 °C for 30 min.
5. **Program and run the end-of-line test (§4, §5) before the cover goes on** (the pogo pads J2 are under the cover).
6. **Rear cover:** press 3 × M2 × 5 mm hex standoffs (F-F) onto the main board with M2 × 4 screws from the front (holes H1–H3), put the cover on (markings outward) and fix it with 3 × M2 × 4 screws. Torque ≈ 0.2 N·m. **No cover = no shipment** (risk H1).
7. Apply the serial/batch label in the S/N field of the cover.
8. **Stand:** push the tree tabs (front row) and the cover tabs (rear row) into the base slots; a drop of cyanoacrylate or a snug press fit (slot width 1.7 mm, tab 1.6 mm).

4. Programming

- Programmer: WCH-LinkE (SWD single-wire), pads J2: `5V – SWIO – GND` (pin 1 = 5V, square pad).
- **Option bytes first: configure NRST as GPIO** – `minichlink -D` (PD7 drives SR_DATA of the shift registers). Without it the lamps light wrongly; nothing is damaged.
- Command: `minichlink -w firmware.bin flash -b` (ch32v003fun) or WCH-LinkUtility.
- Optional: flash the rv003usb bootloader once over SWD (keep RST_MODE = GPIO when its `configurebootloader` sets the option bytes); later firmware updates then go over USB with `minichlink` (HID, no driver).
- After flashing, read back the firmware version / CRC and record it against the serial number.

5. End-of-line test (per unit, ≈ 30 s)

Fixture: USB-C power through a current meter, pogo pins on J2, a light-tight hood with an "eye" photodiode per lamp (or a camera).

Step	Check	Limit
1	Power on at 5.0 V, MCU idle (boost off)	I_in < 10 mA (≈ 1 mA of it is the USB D− pull-up)
2	Firmware **self-test mode** (via SWD flag): ramp HV to 105 V, MCU reports the ADC reading	100–110 V (via ADC); I_in 60–150 mA with all lamps on
3	Each lamp in sequence (camera / photodiodes)	all 10 strike within 1 s, none dark, no wrong lamp (pin-map check)
4	Touch pad: the fixture touches TP2 (through the mask) with a grounded finger probe	firmware registers it
5	Light sensor: hood dark vs. LED on	ADC delta > threshold
6	HV off → discharge	MCU reports < 60 V within 2 s
6b	USB: power from a PC / test host instead of the supply	enumerates as the NeonPine HID device (VID:PID `[…]`), firmware version readable

Step	Check	Limit
7	After the cover is fitted: power on through USB, all lamps light; visual check of lamp bonding, cover screws, label	OK

Log: serial number, firmware CRC, measured currents and HV, and pass/fail → keep with the batch record (traceability, GPSR).

6. Traceability

- Serial / batch format: `NP-1 <rev> <YYWW>-<nnnn>` on the rear-cover label, also in the EOL test log.
- Keep the component lot codes per production batch (lamps especially, plus their radioactivity / RoHS declarations).

7. Manufacturing cost – small series (rev B, prices checked 2026-10-06)

> **Snapshot of the rev B BOM.** The H-bridge + USB board changes the BOM – re-quote before ordering:
>
> | Change since rev B | Effect |
> |---|---|
> | Q11–Q20 MMBTA42 → **MMDT5551** (dual NPN, SOT-363), one per lamp | price ↑ (check) |
> | 10 → **20 × 150k 0603 HV (≥ 200 V)** pull-ups, 22k base resistors 0402, 10 → 20 | count ↑ |
> | + **U2, U3 74HC595BQ** (DHVQFN-16) | new |
> | J1 USB4125-GF-A → **USB4105-GF-A** (USB 2.0, 16P, C3020560) | similar price; the 6-pin cost-down in §7.3 #4 no longer applies |
> | + R13, R14 68R, R15 1k5 (0402), **D4, D5 BZX884-C3V6** (SOD-882, C7529010); C7 removed | ≈ +$0.03 |
> | low-voltage passives 0603 → 0402 | ≈ same |
>
> SMT joints on the back: **277** (rev B: 148). Per finished unit, in **EUR excluding VAT**. USD prices are converted at 1 EUR = 1.15 USD. Assumptions: * **SMT:** JLCPCB "economic" PCBA on the back side. * **Hand work:** lamps, light sensor, cover and stand at ≈ €18/h (CZ). * **Delivery:** to CZ by courier.

7.1 Sources

Item	Source	Basis
Electronic BOM	LCSC / JLCPCB parts library (live query, 2026-10-06)	price break at qty × units; table §7.4
Neon lamps	**TME quote for NEON-4** (EUR incl. 21 % VAT: 10+ 0.287 · 30+ 0.207 · 100+ 0.173 · 600+ 0.129 · 2400+ 0.086)	**Proxy:** the design uses **NEON-3**, same family and size. **Get the NEON-3 quote** – NEON-4 is a 135 V DC type and is not usable on the 105 V rail
SMT assembly	JLCPCB economic PCBA fees: setup $8, stencil $1.5, **$3 per unique extended part**, $0.0016 per joint	148 SMT joints; **12 unique extended parts** → $45.5 per order + $0.24 per board
PCBs	estimate in the style of JLCPCB's calculator (2 layers, 1.6 mm, ENIG, 55 × 69 mm; cover/stand board without copper)	**estimate – run the online quote**
Mechanics, packaging, freight	estimates	

7.2 Unit cost (rev B BOM as designed, with the in-stock substitutes noted in §7.4)

Cost item	10 pcs	100 pcs	500 pcs	Notes
Electronic BOM excl. lamps	2.75	2.15	1.87	$3.17 / $2.47 / $2.15 – see §7.4
10 × neon lamps	1.43	1.07	0.71	TME 100+ / 600+ / 2400+ break, ex VAT (NEON-4 proxy)
Main PCB (ENIG)	2.61	0.35	0.24	≈ $30 / $40 / $140 per order
Cover + stand PCB	1.30	0.39	0.26	≈ $15 / $45 / $150 per order
SMT assembly (JLC)	4.17	0.60	0.29	$45.5 per order + $0.24 per board
Freight PCB + PCBA → CZ	1.74	0.30	0.12	≈ $20 / $35 / $70 per shipment
M2 standoffs ×3 + screws ×6	0.30	0.15	0.12	
Hand assembly	3.60	3.00	1.80	12 / 10 / 6 min (lamp bending jig at 500)
Programming + end-of-line test	0.45	0.36	0.30	1.5 / 1.2 / 1 min
Box, insert, printed manual	1.00	0.70	0.50	
Unit cost, as designed	**≈ 19.4**	**≈ 9.1**	**≈ 6.2**	EUR ex VAT
Unit cost with the cost-down in §7.3	**≈ 18.4**	**≈ 8.3**	**≈ 5.5**	

Import: 21 % CZ import VAT on the JLC shipment (recoverable for VAT payers); PCBs/PCBA are normally duty-free – check the HS code. At 10 pcs the fixed per-order costs dominate: assembly fees, PCB minimums and freight are ≈ €10 of the €19.

7.3 Findings from the price check (act before ordering)

#	Finding	Action
1	**L11 SWPA6028S331MT:** rated only **300 mA** (3.64 Ω) and **0 in stock**	Use **SWPA6045S331MT (C83453)**: 330 µH, 570 mA, 1.27 Ω, $0.05–0.09, in stock. It is 6 × 6 × 4.5 mm with the same land-pattern family and still fits under the 5 mm cover standoffs. Update the value in `tools/gen.py`
2	**C4 1 µF 250 V in 1210 does not exist at LCSC** (1210 only goes to 220 nF at 250 V)	Change the footprint to **1812** and use **C106152** (1 µF 250 V X7R, $0.19–0.38), or use 2 × 470 nF. Small re-place and re-route
3	**Q1 DMN15H310SE: 1 in stock at JLC**, $0.37–0.66	Drop-in SOT-223 alternative **CJT04N15 (C81507)**: 150 V, 160 mΩ at 10 V, V_th 2.5 V, $0.10–0.20, 11 k in stock. **Check R_DS(on) at V_GS = 5 V** on the datasheet. Saves ≈ €0.25/unit
4	**J1** is the most expensive part ($0.58–0.84 for the GCT)	Rev B suggested the 6-pin power-only HC-TYPE-C-6P-01A; **with USB data this is no longer possible.** A cheaper 16P option (e.g. HRO TYPE-C-31-M-12) needs a footprint change and a re-route of the J1 area
5	Q2 TEPT4400 costs $0.31–0.42 (THT, hand-soldered)	Acceptable – it replaces the CdS LDR (RoHS). A cheaper SMD ambient-light sensor would have to go on the front, which means a second SMT side
6	12 extended parts × $3 = $36 per order	Irrelevant at 500 pcs; at 10 pcs it is 30 % of the assembly cost. Using JLC "basic" alternatives where possible (C3 22 µF) saves $3 each

7.4 Electronic BOM detail (LCSC price at the needed quantity, USD per piece)

Ref	Qty	LCSC	Part	Library	10 units	100 units	500 units
U1	1	C5187096	CH32V003F4P6	extended	0.288	0.224	0.163
Q11–Q20	10	C118745	MMBTA42	extended	0.029	0.019	0.017

Ref	Qty	LCSC	Part	Library	10 units	100 units	500 units
Q1	1	C155224	DMN15H310SE-13 (alt. C81507 CJT04N15)	extended	0.535	0.410	0.374
Q2	1	C132178	TEPT4400	extended	0.421	0.341	0.306
L11	1	C83453	SWPA6045S331MT (see 7.3 #1)	extended	0.088	0.072	0.064
D1	1	C85620	ES1D-LTP	extended	0.029	0.029	0.023
D2	1	C283870	SMAJ100A	extended	0.035	0.035	0.028
D3	1	C284108	SMF5.0A	extended	0.019	0.019	0.017
F1	1	C75464	MF-NSMF050-2	extended	0.039	0.039	0.030
J1	1	C3151650	USB4125-GF-A (alt. C2894893)	extended	0.839	0.642	0.584
C4	1	C106152	1 µF 250 V X7R **1812** (see 7.3 #2)	extended	0.302	0.225	0.206
C3	1	C29277	22 µF 10 V X5R 0805	extended	0.082	0.059	0.048
C1	1	C19702	10 µF 10 V 0603	basic	0.032	0.032	0.025
C2, C6, C7	3	C14663	100 nF 0603	basic	0.012	0.012	0.007
C5	1	C57112	10 nF 0603	basic	0.011	0.011	0.009
FB1	1	C1002	600 Ω @ 100 MHz 0603	basic	0.013	0.013	0.010
R21–R30	10	C17470	150 k 0805	basic	0.005	0.004	0.003
R9, R41–R50	11	C31850	22 k 0603	basic	0.004	0.003	0.003
R5, R6	2	C22935	1 M 0603	basic	0.002	0.002	0.002
R3, R4	2	C23186	5.1 k 0603	basic	0.002	0.002	0.001
R1, R2, R7, R11	4	basic	47 / 100 k / 39 k / 1 k 0603	basic	0.002	0.002	0.002
Sum per board					**$3.17**	**$2.47**	**$2.15**
With CJT04N15 + HC-TYPE-C-6P					*$2.04*	*$1.62*	*$1.34*

Lamps: 10 per board → 100 / 1 000 / 5 000 lamps per series. R12 and C8 (snubber) are DNP and not counted.

7.5 One-time costs (not in the unit cost)

Item	≈ EUR
EMC tests (EN 55015 / 61547 + FCC Part 15B), accredited lab	2 600–5 200
Safety evaluation (GPSR file, IEC 60598/62368 method)	900–2 600
WEEE / packaging EPR registrations, authorised representative	250–1 300 per country per year
EOL test fixture (pogo bed, photodiodes) + lamp-bending jig	250–700
Prototype round (rev B samples, 10 pcs)	≈ 200

At 500 units, the one-time costs add **≈ €9–17 per unit**, more than the unit cost itself. Compliance pays off from about 2 000 units (≈ €2.5–4.5 per unit).